

KAOUSHEIK JAYAKUMAR

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College Park, Maryland - 20740

Education

University of Maryland, College Park

August 2025 – May 2027

MS in Computer Science

CGPA: 3.81/4

Advised by Prof. Dinesh Manocha and Prof. Ramani Duraiswami

Indian Institute of Technology Madras

2018 – 2023

Dual Degree (B.Tech + M.Tech) in Electrical Engineering, Minor in Artificial Intelligence

CGPA: 8.59/10

Advised by Prof. S. Umesh

Publications

- TEMPO: Temporally-grounded Multi-task Post-training for Large Audio-Language Models**
Apoorva Kulkarni*, **Kaousheik Jayakumar***, Sreyan Ghosh, Utathya Aich, Ramani Duraiswami, Dinesh Manocha
Under Review
- Playing with Fire : What Transfers When RL Trains a Language Agent?**
Mahesh Ramesh*, **Kaousheik Jayakumar***, Hemanth Ram, Pavan Thodima, Ramani Duraiswami, Dinesh Manocha, Aniket Rege, Emmanouil-Vasileios Vlatakis-Gkaragkounis
Under Review
- Audio-Visual Flamingo: Open Audio-Visual Intelligence for Long and Complex Videos**
Sreyan Ghosh*, Arushi Goel*, **Kaousheik Jayakumar**, Lasha Koroshinadze, Nishit Anand, Zhifeng Kong, Siddharth Gururani, Sang-gil Lee, Jaehyeon Kim, Aya Aljafari, Chao-Han Huck Yang, Sungwon Kim, Ramani Duraiswami, Dinesh Manocha, Mohammad Shoeybi, Bryan Catanzaro, Ming-Yu Liu, Wei Ping
Under Review
- Audio Flamingo Next: Next-Generation Open Audio-Language Models for Speech, Sound, and Music**
Sreyan Ghosh*, Arushi Goel*, **Kaousheik Jayakumar**, Lasha Koroshinadze, Nishit Anand, Zhifeng Kong, Siddharth Gururani, Sang-gil Lee, Jaehyeon Kim, Aya Aljafari, Chao-Han Huck Yang, Sungwon Kim, Ramani Duraiswami, Dinesh Manocha, Mohammad Shoeybi, Bryan Catanzaro, Ming-Yu Liu, Wei Ping
Arxiv preprint - 2026
- Massive Multi-Task Omni Understanding and Reasoning Benchmark for Long Real-World Videos**
Arushi Goel*, Sreyan Ghosh*, Vatsal Agarwal*, Nishit Anand*, **Kaousheik Jayakumar**, Lasha Koroshinadze, Yao Xu, Katie Lyons, James Case, Karan Sapra, Kevin J. Shih, Siddharth Gururani, Abhinav Shrivastava, Ramani Duraiswami, Dinesh Manocha, Andrew Tao, Bryan Catanzaro, Mohammad Shoeybi, Wei Ping
Arxiv preprint - 2026
- A Closer Look at Failure Modes in Temporal Understanding of Large Audio-Language Models**
Apoorva Kulkarni, **Kaousheik Jayakumar**, Sreyan Ghosh, Sarah Wiegrefe, Dinesh Manocha, Ramani Duraiswami
Interspeech - Sydney, 2026
- Do Audio-Visual Large Language Models Really See and Hear?**
Ramaneswaran Selvakumar*, **Kaousheik Jayakumar***, S Sakshi, Sreyan Ghosh, Ruohan Gao, Dinesh Manocha
CVPR Findings – Denver, 2026
- Sparks of Cooperative Reasoning: LLMs as Strategic Hanabi Agents**
Mahesh Ramesh, **Kaousheik Jayakumar**, Aswinkumar Ramkumar, Pavan Thodima, Aniket Rege, Emmanouil-Vasileios Vlatakis-Gkaragkounis
ICML - Seoul, 2026

9. **The Tag-Team Approach: Leveraging CLS and Language Tagging for Enhancing Multilingual ASR**
Kaousheik Jayakumar, Vrunda N. Sukhadia, A Arunkumar, S Umesh
Interspeech – Dublin, 2023
10. **Building Robust and Scalable Multilingual ASR for Indian Languages**
Arjun Gangwar, Kaousheik Jayakumar, S Umesh
Arxiv preprint - 2024

Work Experience

Center of Excellence, Johns Hopkins University

Visiting Graduate Researcher

Baltimore, Maryland

June 2026 – present

- Working on building Audio-Visual retrieval system and multi-video report generation for human annotated claims
- Developing methods to retrieve non-speech audio events from long videos and attribute generated claims back to their source video as citations

Oracle

Member of Technical Staff

Bangalore, Karnataka

July 2023 – Aug 2025

- Integrated edge computing solutions into Oracle Cloud Infrastructure (OCI), unifying architectural components to optimize efficiency, reduce latency, and enhance overall system performance across cloud environments.
- Engineered and executed a large-scale OS upgrade to Oracle Linux 8, ensuring cross-compatibility of dependent components, libraries, and middleware and improved stability of rack-mounted infrastructure.
- Engineered a solution to partition large ISO images into smaller chunks for parallelized security scanning, integrating it back into the infrastructure while ensuring compatibility with existing components, storage, and network pipelines.

Oracle

Product Development Intern

Bangalore, Karnataka

May 2022 – July 2022

- Optimized Docker images by implementing multi-stage builds and minimizing layers, reducing image size and improving deployment speed and runtime efficiency.
- Developed an automated CICD pipeline system that integrated multiple components to generate RPMs and ISO builds
- Automated the configuration management of build environments using Bash/Python scripts, eliminating manual setup overhead and ensuring consistency across development and staging.

Saptang Labs

Software and Cybersecurity Intern

Chennai, Tamil Nadu

Jun 2021 – Jul 2021

- Developed an internal tool to detect vulnerabilities on open ports of any ASN / Organization
- Enhanced machine learning algorithms that were automated into cyber-security solutions for detecting anomalies

Projects and Research Experience

TiDAN: RL-based framework for safety alignment and adaptable Guard-railing of LLMs | *CMSC723*

- Designed and implemented TiDAN, a closed-loop adversarial training framework that co-evolves an AutoDAN-based genetic algorithm attacker with a GRPO-fine-tuned defender, achieving up to 92% reduction in Attack Success Rate on AdvBench (28% → 2.3%) for Qwen3-0.6B across five jailbreak benchmarks (JailBreakBench, InTheWild, AdvBench, StrongREJECT, HarmBench).
- Designed a dual-objective LLM-as-a-Judge reward system that scores safety on malicious prompts and helpfulness on benign prompts (from Tulu-3), preventing over-refusal during alignment. Trained iteratively for 10 GRPO rounds on 4×A6000 GPUs, demonstrating that stronger reward models yield proportionally better safety alignment.
- Extended the framework into an adaptable guardrail pipeline that translates user-defined domain guidelines (e.g., healthcare protocols) into adversarial training signals, demonstrating an 18.5% ASR reduction on HarmBench and 42.8% on AdvBench for domain-specific safety alignment without manual annotation.

Enhancing ASR systems for low-resource languages | *Dual Degree Thesis, IIT Madras*

- Exploited phonemic similarities among various Indian languages and enhanced the performance of multilingual ASR systems for 10 low-resource languages, reducing WER by **2-15%** over the corresponding monolingual baselines.
- Analysed the effect of introducing language ID as an additional learning task which showed further **2-5%** improvement.

National Language Translation Mission (Speech & ASR) | *Bhashini, Govt. of India*

- Developed multilingual Speech systems (ASR, Speech Translation, Synthesis) across multiple Indian languages.
- Curated **4,000 hours** of labelled data from NPTEL lectures for low-resource Indian languages by audio-text alignment.
- Deployed real-time code-switched streaming models for ASR, optimizing inference latency for live-transcription.

Video-to-Video translation pipeline with Lip-Syncing | *IIT Madras*

- Engineered a video-to-video translation pipeline to synchronize lip movements of lecturers with translated audio tracks.
- Implemented and fine-tuned generative models (Wav2Lip/GANs) on technical lecture videos, ensuring temporal consistency and realistic mouth articulation.
- Addressed challenges in pose preservation and face restoration, integrating face detection landmarks to maintain visual fidelity.

Bridge - Multiplayer Co-op Card Game | *Interactive Web-Application*

- Developed a real-time multiplayer application for the classic card game Bridge using React, NodeJs and Socket.IO.

Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, MERN full-stack

Libraries and Frameworks: PyTorch, Tensorflow, Keras, Numpy, Pandas, NodeJS, Django, React, Angular, React Native, NextJS, GraphQL

Databases: Postgres, MySQL, MongoDB

Software and Toolkits: Git, Linux, Shell / Bash scripting, REST API, Docker, Kubernetes, LaTeX, ESPNet, Fairseq

Relevant Coursework: Multimodal Foundational Models, Natural Language Processing (NLP), Probability, Statistics, Computer Vision, Linear Algebra, Data Structures and Algorithms, Estimation Theory, Multi-Arm Bandits

Miscellaneous

Peer Reviewer - ICML 2026	May 2026
Teaching Assistant - Signals and Systems, IIT Madras	Jan 2023 – May 2023
Teaching Assistant - Digital Signal Processing, IIT Madras	July 2022 – Nov 2022
ICPC Regionalist - Secured 120th position out of 721 teams in the Asia Amritapuri regionals	Sept 2020